

Individual Assignment

Moving Average Convergence Divergence (MACD): Construction, Analysis and Use in Decision-Making

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MOVING AVERAGE CONVERGENCE DIVERGENCE (MACD)

Construction, Analysis and Use in Decision-Making

1. Introduction

Moving Average Convergence Divergence (MACD) is a trend-following momentum indicator used in technical analysis. It helps assess whether short-term price momentum is strengthening or weakening relative to a longer-term trend. The standard MACD system contains the MACD Line, Signal Line and MACD Histogram.

2. Construction of MACD

Step 1: Calculate the 12-day and 26-day Exponential Moving Averages (EMAs)

An Exponential Moving Average (EMA) gives greater importance to recent prices. The standard MACD uses a 12-day EMA as the shorter-term average and a 26-day EMA as the longer-term average.

EMA(t) = EMA(previous) + [Close(t) − EMA(previous)] × Multiplier

12-day EMA multiplier = 2/(12+1) = 0.1538; 26-day EMA multiplier = 2/(26+1) = 0.0741.

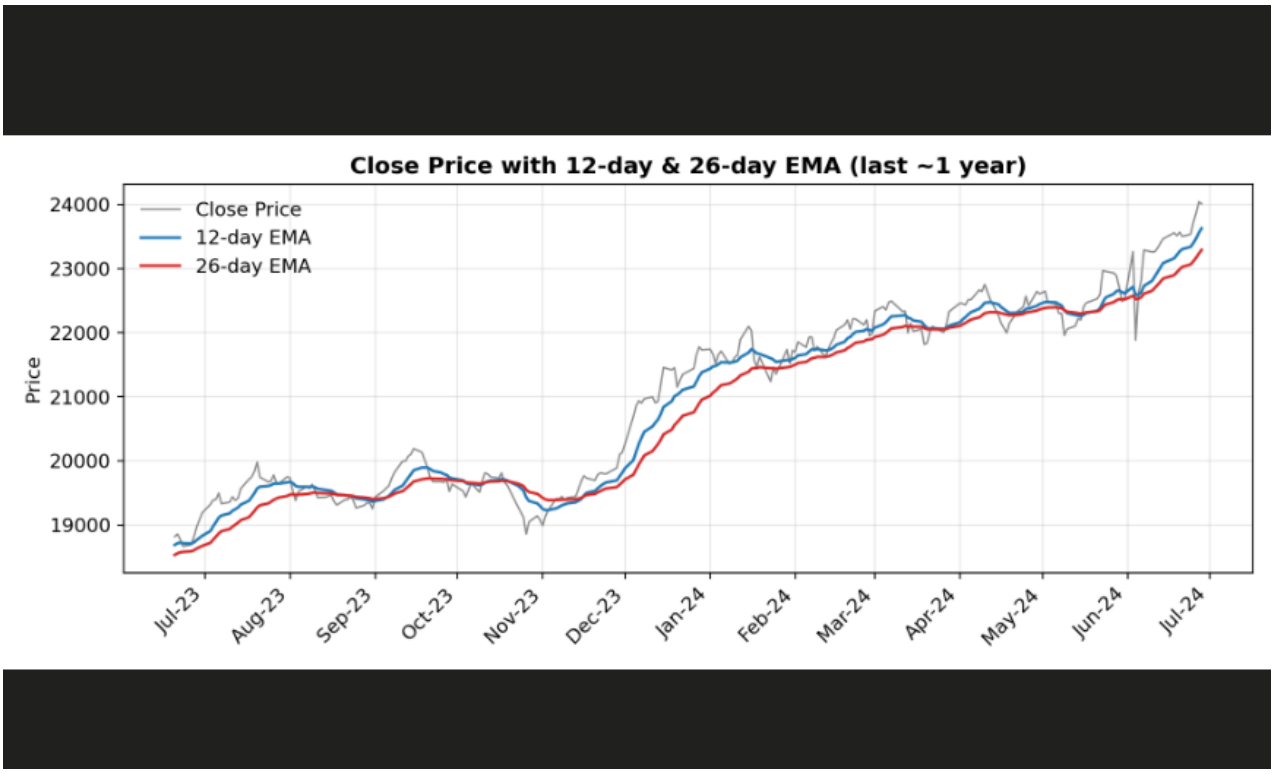


Figure 1. Closing price with 12-day and 26-day EMA — analysis output supplied in the workbook.

Step 2: Calculate the MACD Line

MACD Line = 12-day EMA − 26-day EMA. A positive MACD means the 12-day EMA is above the 26-day EMA; a negative MACD means it is below the 26-day EMA.

MACD — ANALYSIS

3. Signal Line and Histogram

Step 3: Calculate the Signal Line

The Signal Line is a 9-day EMA of the MACD Line itself. The 12-day and 26-day EMAs are calculated from closing prices, while the 9-day EMA is calculated from MACD values.

Signal Line = 9-day EMA of MACD

Step 4: Calculate the MACD Histogram

Histogram = MACD Line − Signal Line. Positive and expanding bars indicate strengthening bullish momentum; shrinking positive bars can indicate weakening bullish momentum. Increasingly negative bars indicate strengthening bearish momentum.

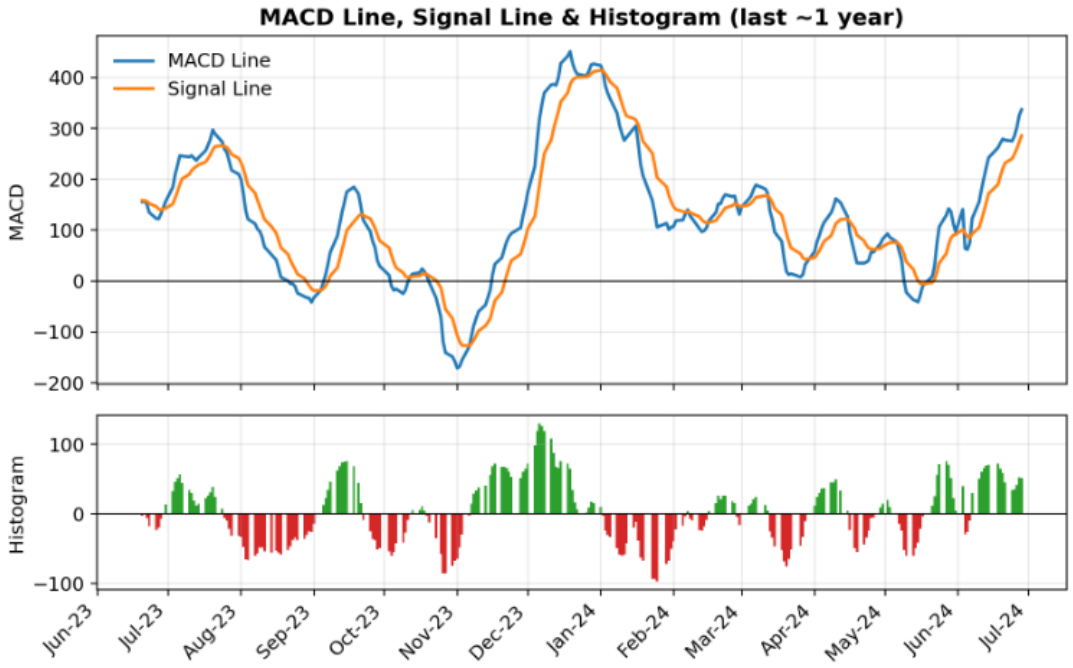


Figure 2. MACD Line, Signal Line and Histogram — analysis output supplied in the workbook.

4. Interpretation of MACD

MACD above zero: the 12-day EMA is above the 26-day EMA. MACD below zero: the 12-day EMA is below the 26-day EMA. MACD crossing above the Signal Line indicates a bullish crossover; crossing below indicates a bearish crossover. Histogram expansion shows increasing momentum, while movement toward zero shows weakening momentum.

Important: A positive MACD should not automatically be treated as a Buy signal, and a negative MACD should not automatically be treated as a Sell signal.

MACD — DECISION-MAKING

5. Use of MACD in Decision-Making

MACD can be used as a momentum and trend-confirmation tool. A practical process is to observe the MACD, identify the crossover or zero-line position, assess histogram momentum, confirm with price action and other evidence, and then decide.

6. Observations from the Analysis

25 Jul 2023: Bearish crossover. Close = 19,680.60; MACD = 259.15; Signal Line = 264.82. Interpretation: momentum turned relatively bearish.

07 Nov 2023: Bullish crossover. Close = 19,406.70; MACD = -108.20; Signal Line = -123.15. Interpretation: momentum turned relatively bullish.

07 Jun 2024: Bullish crossover. Close = 23,290.15; MACD = 122.88; Signal Line = 92.75. Interpretation: momentum turned relatively bullish.

7. Worked Example from the Analysis Workbook

For 27 June 2024: Close = 24,044.50; 12-day EMA = 23,559.41; 26-day EMA = 23,234.49. MACD = 324.91; Signal Line = 272.93; Histogram = 51.99. The positive MACD and positive Histogram support a relatively positive momentum interpretation, but do not by themselves establish a Buy decision.

8. Limitations

MACD is a lagging (reactive) indicator because it is based on moving averages. It can produce false signals, especially in sideways or volatile markets. It should therefore be combined with price action, volume, support/resistance, and where appropriate, fundamental analysis.

9. Conclusion

MACD evaluates trend and momentum by comparing a 12-day EMA with a 26-day EMA, smoothing the MACD Line through a 9-day Signal Line, and measuring their difference through the Histogram. Its key value is in identifying changes in momentum and confirming trend conditions rather than making isolated Buy or Sell decisions.

Sources

1. Course reading: Moving Average Convergence Divergence (MACD), provided for the assignment. 2. MACD analysis workbook supplied for the assignment; calculations and figures reproduced from the student's analysis output.
2. https://docs.google.com/spreadsheets/d/1JFsspJPoiuo1QF3Ave_NRmJCksnsghv_/edit?usp=sharing&ouid=113182242353459829859&rtpof=true&sd=true